Div & curl in 3D (16.5)

$$\vec{F} = \langle P, Q, R \rangle = P\hat{i} + Q\hat{j} + R\hat{k}$$

$$\boxed{\operatorname{div} \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}}$$

$$\boxed{\begin{aligned} \operatorname{curl} \vec{F} = &\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\hat{i} + \\ &+ \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{k} \end{aligned}}$$

Think of ∇ as a "vector":

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

Then: $\boxed{\begin{aligned} \operatorname{div} \vec{F} &= \nabla \cdot \vec{F} \\ \operatorname{curl} \vec{F} &= \nabla \times \vec{F} \end{aligned}}$

Example: $\vec{F} = xz\,\hat{i} + xyz\,\hat{j} - y^2\,\hat{k}$

$$\operatorname{curl}(\vec{F}) = \nabla \times \vec{F} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xz & xyz & -y^2 \end{bmatrix}$$

$$= (-2y - xy)\,\hat{i} - (0 - x)\,\hat{j} + (yz - 0)\,\hat{k}$$

Fluid flow interpretation

V = small cube of fluid

$\vec{F}$ = velocity of V

$$\nabla \cdot \vec{F} = \frac{d}{dt}(\text{volume of } V)$$

$$\nabla \times \vec{F} = 2\begin{pmatrix} \text{angular velocity} \\ \text{vector of } V \end{pmatrix}$$

Electromagnetism

Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

Maxwell's equations:

$$\nabla \cdot \vec{E} = \rho/\varepsilon_0$$
$$\nabla \times \vec{E} = -\partial \vec{B}/\partial t$$
$$\nabla \cdot \vec{B} = 0$$
$$c^2 \nabla \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \vec{j}/\varepsilon_0$$

Identities:

1) $\nabla \cdot (\nabla f) = \nabla \cdot \langle f_x, f_y, f_z \rangle$
$= f_{xx} + f_{yy} + f_{zz} =: \nabla^2 f$
"Laplacian"

2) $\nabla \times (\nabla f) = \langle 0, 0, 0 \rangle$

3) $\nabla \cdot (\nabla \times \vec{F}) = 0$

Warning: $\nabla \times (\overbrace{\nabla \cdot \vec{F}}^{\text{function}})$ does not make sense!

Proof:

2) $\nabla \times \langle f_x, f_y, f_z \rangle =$

$$= \det \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \end{vmatrix} =$$

$$= \hat{i}\left(\cancel{f_{zy} - f_{yz}}\right) + \hat{j}\left(\cancel{f_{xz} - f_{zx}}\right) + \hat{k}\left(\cancel{f_{yx} - f_{xy}}\right)$$

3) $\nabla \cdot (\nabla \times \langle P, Q, R \rangle) =$

$$= \nabla \cdot \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$$

$$= \cancel{R_{yx}} - \cancel{Q_{zx}} + \cancel{P_{zy}} - \cancel{R_{xy}} + \cancel{Q_{xz}} - \cancel{P_{yz}}$$

Theorem: $\vec{F}$ vector field on $\mathbb{R}^3$

$\nabla \times \vec{F} = \vec{0} \iff$ there exists a function f such that $\vec{F} = \nabla f$

$\nabla \cdot \vec{F} = 0 \iff$ there exists a vector field $\vec{A}$ such that $\vec{F} = \nabla \times \vec{A}$.